CMB-Calibrated Geometric Measurement of Void-Node Structure

A Dual Null-Probe Framework for Quantifying Path-Integrated Redshift in Curvature-Minimum Spacetime

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Abstract

Mass Deficit Nodes (MDNs) have been identified as curvature-minimum spacetime structures organizing expansion-dominated regions of the cosmic web. Previous work established that photon propagation through MDN-organized geometry produces a non-canceling, path-integrated contribution to cosmological redshift arising from null–timelike congruence decoupling. Paper 5 introduced a depth-slicing method that isolates these geometric contributions within finite spacetime intervals through differential line-of-sight measurements.

In the present work, we introduce a complementary framework that uses the cosmic microwave background (CMB) as a **line-of-sight geometric calibrator**. Because both optical photons and CMB photons propagate along null geodesics through the same intervening spacetime structure, they encode consistent information about the geometry of that path. We show that CMB temperature residuals provide an independent observable correlated with MDN basin structure, enabling calibration of geometric redshift contributions.

This approach does not attempt full decomposition of cosmological redshift into kinematic and geometric components. Instead, it provides a means of **calibrating and validating the geometric signal isolated via depth slicing**, yielding a dual null-probe framework for testing curvature-minimum spacetime structure using existing data.

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1. Introduction

Cosmological observations rely fundamentally on null propagation through spacetime. In the standard interpretation, redshift is treated primarily as a kinematic observable associated with global expansion. However, previous work has established that in an inhomogeneous universe organized by curvature extrema, photon propagation accumulates a **path-integrated geometric contribution** driven by the structure of spacetime itself.

In Papers 1–4, we identified Mass Deficit Nodes (MDNs) as curvature minima organizing cosmic voids and derived the associated redshift accumulation through null congruence evolution. In Paper 5, we introduced a **depth-slicing method** that isolates this geometric contribution within discrete intervals by exploiting exact cancellation of shared foreground structure.

The purpose of the present work is to introduce a **second, independent observational probe** of the same geometry by leveraging the cosmic microwave background.

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2. Dual Null Probes of Spacetime Geometry

Photons from distant galaxies and photons of the CMB both propagate along null geodesics governed by the same underlying spacetime metric.

Thus:

> Null geodesics encode geometry independent of photon origin or energy.

Optical photons accumulate geometric redshift:

$$\Delta z_{\text{MDN}} = \frac{1}{3} \int \Theta_{\text{eff}}(\lambda) \, d\lambda$$

CMB photons acquire temperature anisotropies along the same paths:

$$\frac{\Delta T}{T} = \left(\frac{\Delta T}{T}\right)_{\text{primordial}} + \left(\frac{\Delta T}{T}\right)_{\text{prop}}$$

where the propagation term reflects intervening structure.

Key Principle

> Both observables encode projections of the same line-of-sight geometry.

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3. Geometric Redshift from MDN Basin Traversal

From the explicit solution derived in Paper 4, the contribution from a single MDN basin is:

$$\Delta z_{\mathrm{MDN}} = \frac{2}{3}\ln\left(1+\frac{1}{2}\theta_0 L\right)$$

For a general line of sight:

$$\Delta z_{\mathrm{MDN,\,total}} = \sum_i \frac{2}{3}\ln\left(1+\frac{1}{2}\theta_{0,i} L_i\right)$$

This expression depends on:

- basin expansion strength $\theta_{0,i}$
- affine traversal length L_i
- cumulative void structure along the path

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4. CMB Temperature Residuals as Geometric Probes

Observed CMB temperature fluctuations along a given direction may be written schematically as:

$$\frac{\Delta T}{T} = \left(\frac{\Delta T}{T}\right)_{\mathrm{primordial}} + \left(\frac{\Delta T}{T}\right)_{\mathrm{ISW}} + \left(\frac{\Delta T}{T}\right)_{\mathrm{geom}}$$

where:

- the ISW term reflects time-varying potentials
- the geometric term reflects integrated spacetime structure

While the decomposition is not uniquely separable observationally, the total residual contains contributions from:

- void depth
- expansion profile
- path length through underdense regions

Interpretation

$$\frac{\Delta T}{T}_{\text{LOS}} \iff \text{integrated void-node geometry}$$

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5. Transfer Relation Between Observables

Because both redshift accumulation and CMB residuals depend on the same underlying geometry, we define an effective relation:

$$\Delta z_{\text{MDN}} = F \cdot \left(\frac{\Delta T}{T}\right)_{\text{LOS}}$$

where F is a dimensionless transfer function.

Properties of F

- depends on large-scale geometric structure
- varies slowly across similar environments
- encodes mapping between thermal and optical observables

To leading order, both observables scale with:

- cumulative basin exposure
- relative expansion enhancement
- void filling fraction

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6. Role of CMB as a Calibrator

Unlike earlier formulations, we do not interpret this relation as enabling **complete decomposition** of observed redshift.

Instead:

CMB provides an independent calibration of geometric signal strength.

Specifically, it allows:

- estimation of relative basin strength along a line of sight
- validation of depth-sliced measurements
- normalization of geometric contributions

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7. Integration with Depth Slicing

From Paper 5, the measurable quantity is:

$$\delta z_{1,2} = \sum_{i \in \text{shell}} \Delta\, z_{\text{MDN},i}$$

The CMB provides a corresponding measure:

$$\left(\frac{\Delta T}{T}\right)_{\text{shell}}$$

Along the same angular direction.

Combined Interpretation

$$\delta z_{1,2} \iff F \cdot \left(\frac{\Delta T}{T}\right)_{\text{shell}}$$

This allows:

- cross-correlation of independent observables
- calibration of basin-level contributions
- consistency testing of geometric predictions

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8. Observational Protocol

Step 1
Select angularly localized sightlines from Paper 5 analysis

Step 2
Compute depth-sliced redshift residuals $\delta z_{1,2}$

Step 3
Extract corresponding CMB temperature along same directions

Step 4
Compare:

$$\delta z_{1,2} \text{vs} \left(\frac{\Delta T}{T}\right)_{\text{LOS}}$$

Step 5
Fit or constrain transfer relation F

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9. Testable Predictions

The framework yields three key predictions:

9.1 Correlation

$$\delta z_{1,2} \text{ correlates with } \frac{\Delta T}{T}_{\text{LOS}}$$

9.2 Consistency Across Scales

The inferred F remains approximately stable across comparable structures.

9.3 Joint Reduction of Residual Scatter

Combining both observables reduces unexplained variance compared to either alone.

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10. Distinction from Previous Approaches

Earlier proposals attempted:

- direct subtraction of geometric components
- full decomposition of redshift

The present framework:

- avoids full decomposition
- treats geometric contribution as **measurable but cumulative**
- uses independent observables for validation

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11. Limitations

- separation of ISW and geometric components is incomplete
- noise in CMB data limits small-scale resolution
- transfer function F requires empirical calibration

These limitations affect precision, not principle.

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12. Discussion

The use of dual null probes reframes the observational problem. Instead of attempting to isolate geometry from a single dataset, we leverage:

- optical redshift (path-integrated expansion)
- CMB temperature (path-integrated geometry)

This introduces a fundamentally new approach:

probing spacetime structure through **multi-channel null observables**

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13. Conclusion

We have introduced a CMB-based calibration framework that complements depth-sliced measurements of geometric redshift.

The method:

- uses the CMB as an independent probe of line-of-sight geometry
- calibrates MDN-induced redshift contributions
- enables cross-validation of geometric effects

Together with depth slicing, this establishes a **dual null-probe observational program** for testing curvature-minimum spacetime structure within general relativity.